Exercise 2 - SQL Fundamentals (Aggregate Functions & Grouping)

employees

id	first_name	last_name	department	salary	hire_date	city
1	John	Doe	IT	55 000	2018-06-15	New York
2	Jane	Smith	HR	48 000	2019-07-20	Chicago
3	Mike	Johnson	finance	60 000	2017-09-30	Los Angeles
4	Sarah	Brown	IT	53 000	2021-03-25	New York
5	David	White	Marketing	52 000	2016-04-10	San Francisco
6	Emily	Davis	IT	62 000	2015-02-14	Chicago
7	Robert	Wilson	Finance	59 000	2019-10-01	Houston
8	Jessica	Moore	HR	51 000	2018-05-22	Los Angeles
9	Daniel	Clark	Marketing	53 000	2022-06-01	Chicago
10	Laura	Hall	IT	50 000	2020-08-10	San Francisco

1. COUNT() Function

Q - Write a SQL query to find the total number of employees in the company

A - SELECT (*) AS total_number_of_employees
FROM employees;

total_number_of_employees
10

2. SUM() Function

Q - Write a SQL query to find the total salary paid to all employees in the IT department.

A - SELECT Sum(Salary) As total_salary
FROM employees
WHERE department = ('IT');

total_salary
220 000

3. AVG() Function

Q - Write a SQL query to calculate the average salary of employees in the HR dep.

A - SELECT AVG(salary) As average_salary
FROM employees
WHERE department = ('HR'); →

average_salary
49 500

4. MIN() and MAX() Function

Q- Write a SQL query to find the highest and lowest salary in the company

A- SELECT MIN(salary) AS lowest_salary,
MAX (salary) AS highest_salary
FROM employees ;

lowest-salary	highest-salary
48 000	62 000

5. GROUP BY Statement

Q- Write a SQL query to group employees by department and display the total salary paid in each department.

A- SELECT Sum(salary) AS total_salary,
department
FROM employees
GROUP BY department ;

total-salary	department
220 000	IT
99 000	HR
119 000	Finance
105 000	Marketing

6. GROUP BY and COUNT ()

Q= Write a SQL query to count how many employees work in each city

A- SELECT COUNT (*) AS employees,
city
FROM employees
GROUP BY city ;

employees	city
2	New York
3	Chicago
2	Los Angeles
2	San Francisco
1	Houston

7. GROUP BY and ORDER BY

Q. Write a SQL query to group employees by department, calculate the average salary in each department, and order the results in descending order of average salary.

A. SELECT department,
    AVG (Salary) AS average_salary
FROM employees
GROUP BY department
ORDER BY average_salary DESC;

department	average_salary
Finance	54 500
IT	55 000
marketing	52 500
HR	49 500

8. HAVING Clause

Q. Write a SQL query to find departments where the total salary paid exceeds 100 000.

A. SELECT department,
    Sum (salary) AS total_salary_paid
FROM employees
GROUP BY department
HAVING Sum (salary) > 100000;

department	total_salary_paid
IT	220 000
Finance	119 000
Marketing	105 000

9. Combining GROUP BY, HAVING, and ORDER BY

Q- Write a SQL query to list cities where more than one employee works, ordered by the number of employees in descending order.

A- SELECT COUNT (*) AS number_of_employees,
city

FROM employees
GROUP BY city

ORDER BY number_of_employees DESC ;

number_of_employees	city
3	Chicago
2	New York
2	San Francisco
2	Los Angeles
1	Houston

10. Combining Aggregate Functions

Q- Write a SQL query to find the department with the highest average salary.

A- SELECT department
AVG (salary) AS highest_average_salary

FROM employees
GROUP BY department ;
ORDER By highest_average_salary DESC
LIMIT 1 ;

department	highest_average_salary
Finance	59 500